

# Physical Chemistry (I)

## Lecture 06



# In the Last Class

- Internal energy of a perfect gas
  - Maxwell-Boltzmann distribution
  - Equipartition theorem
- The first law of thermodynamics
  - The first law
  - Work
  - Heat



# Internal Energy of Perfect Gas

$$U = \frac{1}{2} M n \langle v^2 \rangle$$

From the Maxwell-Boltzmann distribution,

$$\langle v^2 \rangle = v_{\text{rms}}^2 = \frac{3RT}{M}$$

Hence,

$$U = \frac{3}{2} nRT$$

# Equipartition Theorem

For a sample at thermal equilibrium,  
the average value of each quadratic contribution  
to the internal energy is  $(1/2)k_B T$ .

(Quadratic contribution:  $E \propto p_x^2$  or  $E \propto x^2$ )

# First Law of Thermodynamics

“The internal energy of an isolated system is constant.”

- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$dU = \delta q + \delta w$$

↑                      ↑                      ↑  
small change of  $U$     small  $q$             small  $w$

# Different Types of Work

$$\delta w = (\text{intensive factor}) \times (\text{extensive factor})$$

| Type of work      | $dw$               | Comments                                 | Units <sup>†</sup> |
|-------------------|--------------------|------------------------------------------|--------------------|
| Expansion         | $-p_{\text{ex}}dV$ | $p_{\text{ex}}$ is the external pressure | Pa                 |
|                   |                    | $dV$ is the change in volume             | $\text{m}^3$       |
| Surface expansion | $\gamma d\sigma$   | $\gamma$ is the surface tension          | $\text{N m}^{-1}$  |
|                   |                    | $d\sigma$ is the change in area          | $\text{m}^2$       |
| Extension         | $f dl$             | $f$ is the tension                       | N                  |
|                   |                    | $dl$ is the change in length             | m                  |
| Electrical        | $\phi dQ$          | $\phi$ is the electric potential         | V                  |
|                   |                    | $dQ$ is the change in charge             | C                  |
|                   | $Q d\phi$          | $d\phi$ is the potential difference      | V                  |
|                   |                    | $Q$ is the charge transferred            | C                  |



# Calculation of Expansion Work

$$\delta w = -p_{\text{ex}} dV$$

- Total work for a change from **state  $i$**  to **state  $f$**

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} (-p_{\text{ex}} dV) = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

1. Expansion against constant external pressure
2. Reversible expansion

# Heat

$$\delta q = C dT$$

- Total heat for a change from state  $i$  to state  $f$

$$q = \int_{\text{state } i}^{\text{state } f} \delta q = \int_{\text{state } i}^{\text{state } f} (C dT) = \int_{T_i}^{T_f} C dT$$

# First Law Revisited

$$dU = \delta q + \delta w$$

$$dU = \delta q + \delta w_{\text{exp}} + \delta w_{\text{add}}$$

$$dU = C dT - p_{\text{ex}} dV + \delta w_{\text{add}}$$

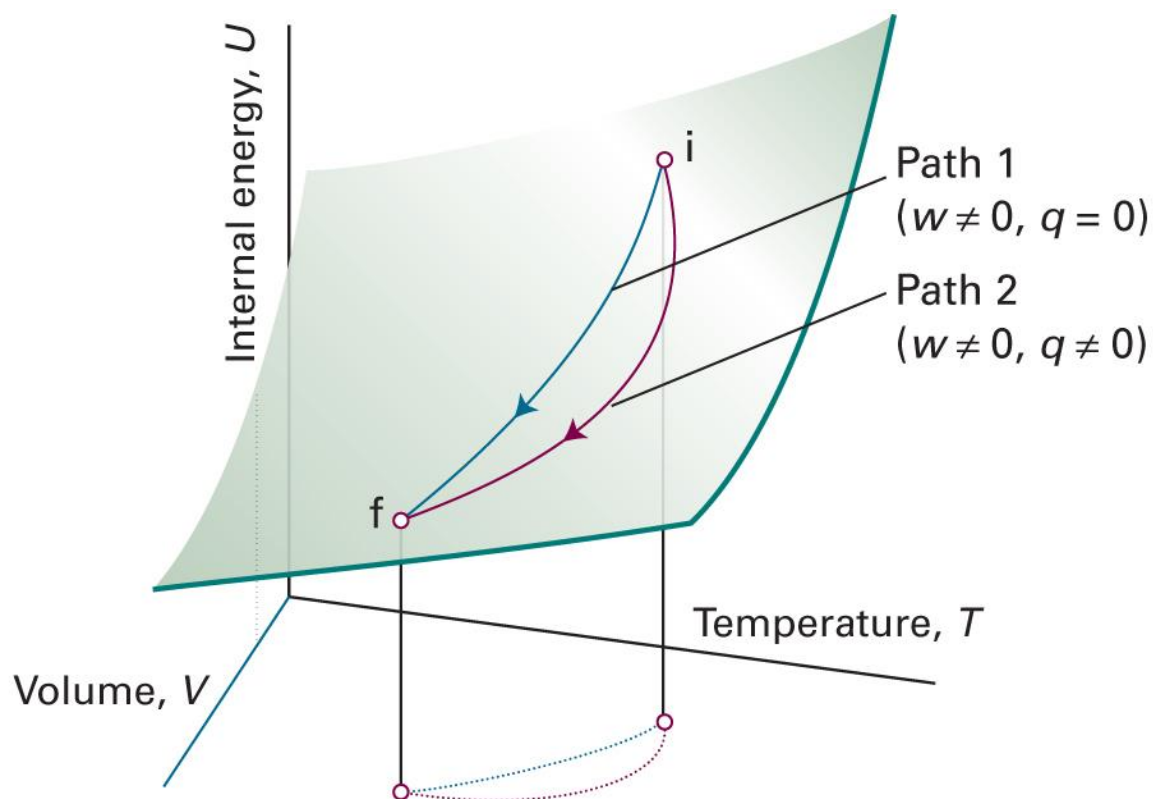
For  $\delta w_{\text{add}} = 0$ ,

$$dU = C dT - p_{\text{ex}} dV$$

# State and Path Functions

- **State function:** a property that depends only on the current state of a system and is independent of its history  
e.g. internal energy, pressure, volume, temperature
- **Path function:** a property with a value that depends on the path between two states  
e.g. heat, work

# State and Path Functions



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11<sup>th</sup> ed., Figure 2D.1)

# Related Mathematics

- **Exact differential:** an infinitesimal quantity that, when integrated, gives a result that is independent of the path between the initial and final states.

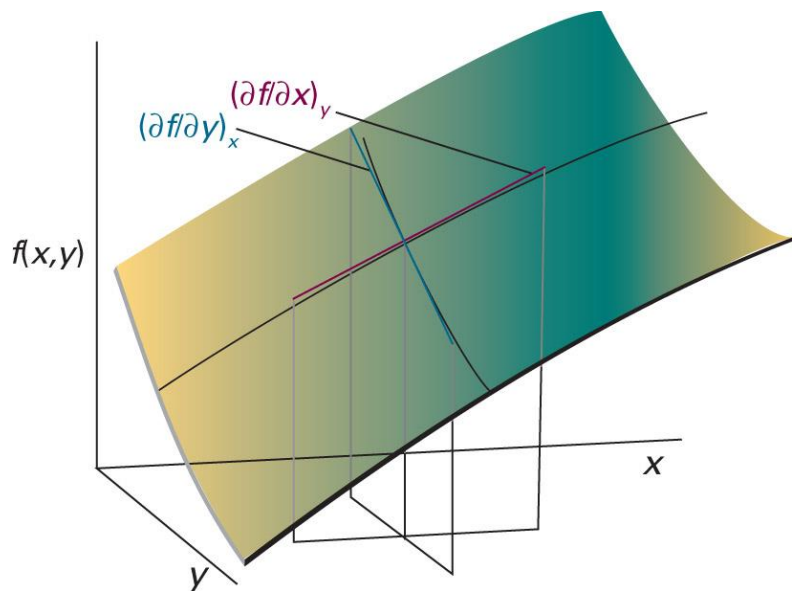
$$\Delta U = \int_{\text{state } i}^{\text{state } f} dU$$

- **Inexact differential:** an infinitesimal quantity that, when integrated, gives a result that depends on the path between the initial and final states.

$$q = \int_{\text{state } i, \text{ path}}^{\text{state } f} \delta q$$

# Partial Derivatives

“the slope of the function with respect to one of the variables, all the other variables being held constant”



$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

# Differential

If  $f$  is a function of  $x$  and  $y$ , the differential  $df$  is

$$df = \left( \frac{\partial f}{\partial x} \right)_y dx + \left( \frac{\partial f}{\partial y} \right)_x dy$$

when  $df$  is an **exact differential**.

# State Variables

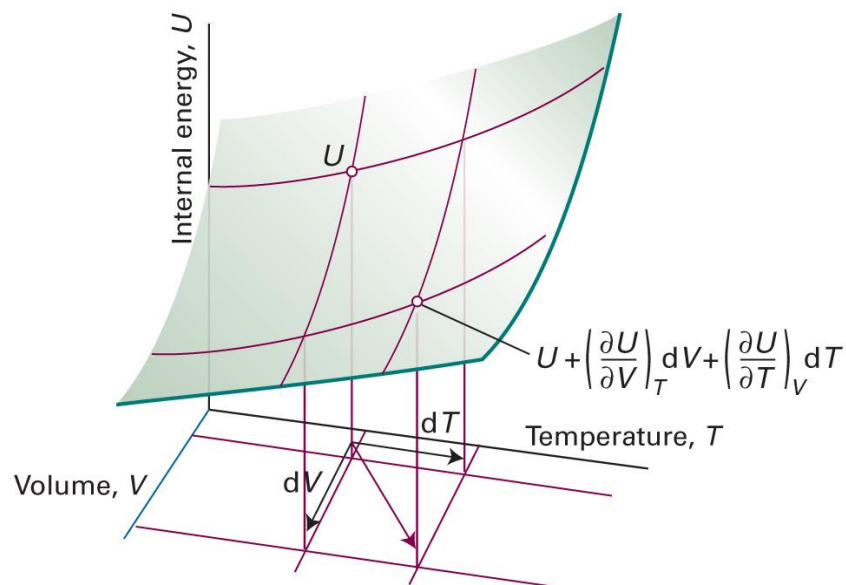
- A state is defined by  $p, V, n, T$
- Typically  $n$  is fixed
- Equation of state:  $p$  is constrained by  $V$  and  $T$
- We can just use  $V$  and  $T$  to define a state

$$U = U(V, T)$$

# Changes in Internal Energy

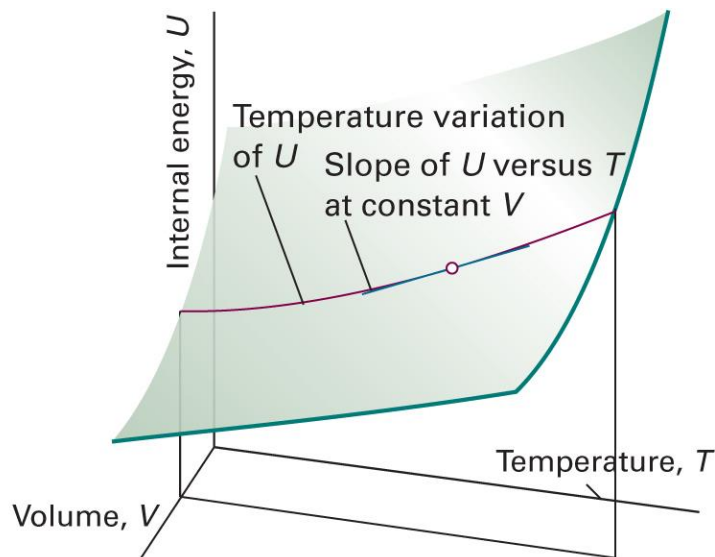
For  $U = U(V, T)$ ,

$$dU = \left( \frac{\partial U}{\partial V} \right)_T dV + \left( \frac{\partial U}{\partial T} \right)_V dT$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11<sup>th</sup> ed., Figure 2D.2)

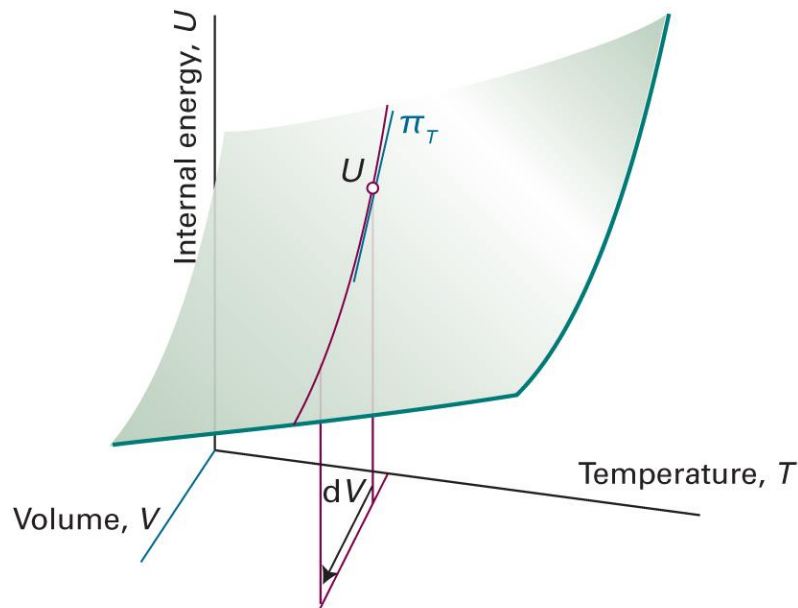
# Meaning of $(\partial U / \partial T)_V$



$$dU = C_V dT \text{ (at constant } V\text{)}$$

$$\left( \frac{\partial U}{\partial T} \right)_V = C_V$$

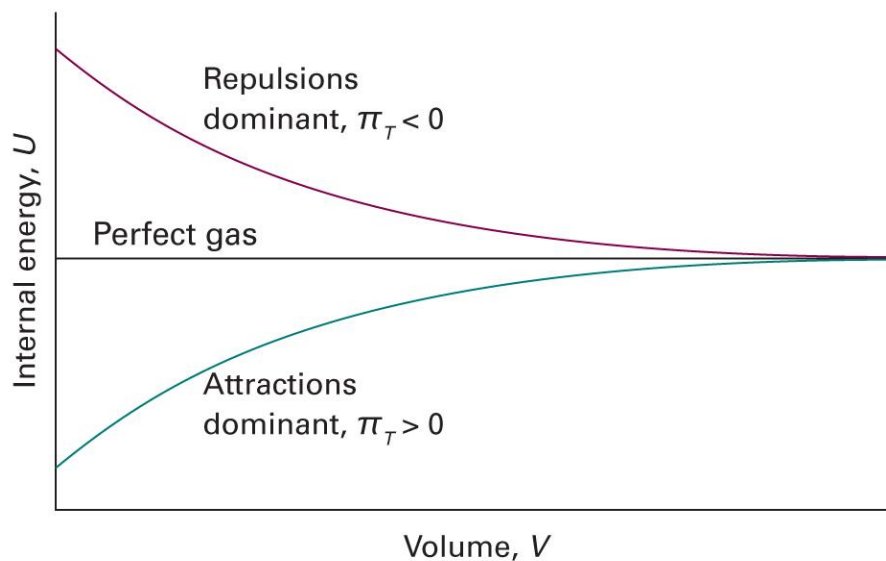
# Meaning of $(\partial U / \partial V)_T$



variation of the internal energy of a substance  
as its volume is changed at constant temperature

# Internal Pressure

$$\pi_T = \left( \frac{\partial U}{\partial V} \right)_T$$



# Use of Differentials

$$dU = \underbrace{\left(\frac{\partial U}{\partial V}\right)_T}_{\pi_T} dV + \underbrace{\left(\frac{\partial U}{\partial T}\right)_V}_{C_V} dT$$

$$dU = \pi_T dV + C_V dT$$

At constant pressure,

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

# Response Functions

- Expansion coefficient

$$\alpha = \frac{1}{V} \left( \frac{\partial V}{\partial T} \right)_p$$

- Isothermal compressibility

$$\kappa_T = -\frac{1}{V} \left( \frac{\partial V}{\partial p} \right)_T$$

# Some Data

|                 | $\alpha / (10^{-4} \text{ K}^{-1})$ | $\kappa_T / (10^{-6} \text{ bar}^{-1})$ |
|-----------------|-------------------------------------|-----------------------------------------|
| <i>Liquids:</i> |                                     |                                         |
| Benzene         | 12.4                                | 90.9                                    |
| Water           | 2.1                                 | 49.0                                    |
| <i>Solids:</i>  |                                     |                                         |
| Diamond         | 0.030                               | 0.185                                   |
| Lead            | 0.861                               | 2.18                                    |

# General Expression

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

Using the definition of the expression coefficient  $\alpha$ ,

$$\left(\frac{\partial U}{\partial T}\right)_p = \alpha \pi_T V + C_V$$

For a perfect gas,  $\pi_T = 0$  and

$$\left(\frac{\partial U}{\partial T}\right)_p = C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

# Example 2D.2

Derive an expression for the expansion coefficient of a perfect gas.

$$\alpha = \frac{1}{V} \left( \frac{\partial V}{\partial T} \right)_p$$

For a perfect gas,  $V = nRT/p$

$$\left( \frac{\partial V}{\partial T} \right)_p = \left( \frac{\partial}{\partial T} \left( \frac{nRT}{p} \right) \right)_p = \frac{nR}{p}$$

Thus,

$$\alpha = \frac{1}{V} \frac{nR}{p} = \frac{nR}{nRT} = \frac{1}{T}$$

# Summary for Chapter 3 (1)

First Law of Thermodynamics

$$dU = \delta q + \delta w$$

Work

$$\delta w = -p_{\text{ex}}dV + \delta w_{\text{add}}$$

- Constant- $p$  change
- Reversible change

Heat

$$\delta q = C dT$$

- Constant- $V$  change

$$C_V = (\partial U / \partial T)_V$$

Thermodynamics

Mathematics

State function

Exact differential

Path function

Inexact differential

Differential form of  $U$

$$dU = \left( \frac{\partial U}{\partial V} \right)_T dV + \left( \frac{\partial U}{\partial T} \right)_V dT$$

- Internal pressure  $\pi_T$
- Response functions  $\alpha$ ,  $\kappa_T$

# Enthalpy $H$

$$H = U + pV$$

- Energy required to create a system ( $U$ ) and to make a room for the system ( $pV$ )
- Since  $U$ ,  $p$ , and  $V$  are state functions,  $H$  is also a state function
- Molar enthalpy  $H_m = H/n$

# Example 2B.1

The change in molar internal energy when  $\text{CaCO}_3(\text{s})$  as calcite converts to its polymorph aragonite, is  $+0.21 \text{ kJ mol}^{-1}$ . Calculate the difference between the molar enthalpy and internal energy changes when the pressure is 1.0 bar. The mass densities of the polymorphs are  $2.71 \text{ g cm}^{-3}$  (calcite) and  $2.93 \text{ g cm}^{-3}$  (aragonite).

The enthalpy change is

$$\begin{aligned}\Delta H_m &= H_m(\text{aragonite}) - H_m(\text{calcite}) \\ &= [U_m(\text{a}) + pV_m(\text{a})] - [U_m(\text{c}) + pV_m(\text{c})] \\ &= \Delta U_m + p[V_m(\text{a}) - V_m(\text{c})]\end{aligned}$$

## Example 2B.1

$$\Delta H_m = \Delta U_m + p[V_m(a) - V_m(c)]$$

Since  $V_m = M/\rho$ ,

$$\Delta H_m = \Delta U_m + pM \left[ \frac{1}{\rho(a)} - \frac{1}{\rho(c)} \right]$$

Substitution of data provides

$$\begin{aligned} \Delta H_m - \Delta U_m &= (1.0 \times 10^5 \text{ Pa}) \times (100.09 \text{ g mol}^{-1}) \times [1/(2.93 \text{ g cm}^{-3}) - 1/(2.71 \text{ g cm}^{-3})] \\ &= -2.8 \times 10^5 \text{ Pa cm}^3 \text{ mol}^{-1} = -0.28 \text{ J mol}^{-1} \end{aligned}$$

# Gas-Involving Processes

$$H = U + pV$$

For a perfect gas,

$$H = U + nRT$$

that is,

$$\Delta H \approx \Delta U + RT\Delta n_g$$

# Infinitesimal Change

For an infinitesimal change in the state,

$$U \rightarrow U + dU$$

$$p \rightarrow p + dp$$

$$V \rightarrow V + dV$$

$$H \rightarrow H + dH$$

From  $H = U + pV$ ,

$$H + dH = (U + dU) + (p + dp)(V + dV)$$

# Infinitesimal Change in $H$

$$H + dH = (U + dU) + (p + dp)(V + dV)$$

$$H + dH = U + dU + pV + pdV + Vdp + \cancel{dpdV}$$

too small

Since  $H = U + pV$ ,

$$dH = dU + pdV + Vdp$$

From the first law of thermodynamics,

$$dH = \delta q + \delta w + pdV + Vdp$$

# Reversible Isobaric Process

$$dH = \delta q + \delta w + pdV + Vdp$$

For a reversible change,

$$\delta w = -p_{\text{ex}}dV = -pdV$$

so that

$$dH = \delta q - pdV + pdV + Vdp = \delta q + Vdp$$

At constant pressure ( $dp = 0$ ),

$$dH = \delta q_p$$

# Enthalpy Change

$$dH = \delta q_p$$

Integration leads to

$$\Delta H = \int_{\text{state } i}^{\text{state } f} dH = \int_{\text{state } i}^{\text{state } f} \delta q_p = q_p$$

# Heat Capacity at Constant $p$

Since  $\delta q = C dT$ ,

$$dH = \delta q_p = C_p dT$$

which means

$$C_p = \left( \frac{\partial H}{\partial T} \right)_p$$

The change in enthalpy is

$$\Delta H = \int_{\text{state } i}^{\text{state } f} dH = \int_{\text{state } i}^{\text{state } f} (C_p dT) = \int_{T_i}^{T_f} C_p dT$$

# At Constant $p$

$$\Delta H = \int_{T_i}^{T_f} C_p dT$$

If  $C_p$  is constant,

$$\Delta H = \int_{T_i}^{T_f} C_p dT = C_p \int_{T_i}^{T_f} dT = C_p(T_f - T_i) = C_p\Delta T$$

Since  $\Delta H = q_p$  at constant pressure,

$$q_p = C_p\Delta T$$

# $T$ -Dependence of $C_{p,m}$

Empirical expression:

$$C_{p,m} = a + bT + \frac{c}{T^2}$$

|                     | $a / (\text{J K}^{-1} \text{mol}^{-1})$ | $b / (10^{-3} \text{J K}^{-2} \text{mol}^{-1})$ | $c / (10^5 \text{J K mol}^{-1})$ |
|---------------------|-----------------------------------------|-------------------------------------------------|----------------------------------|
| C(s, graphite)      | 16.86                                   | 4.77                                            | -8.54                            |
| CO <sub>2</sub> (g) | 44.22                                   | 8.79                                            | -8.62                            |
| H <sub>2</sub> O(l) | 75.29                                   | 0                                               | 0                                |
| N <sub>2</sub> (g)  | 28.58                                   | 3.77                                            | -0.50                            |

## Example 2B.2

What is the change in molar enthalpy of  $\text{N}_2$  when it is heated from  $25^\circ\text{C}$  to  $100^\circ\text{C}$ ? Use the heat capacity information in the previous table.

$$\Delta H = \int_{T_i}^{T_f} C_p dT$$

$$C_{p,m} = a + bT + \frac{c}{T^2}$$

Hence,

$$\Delta H_m = \int_{T_i}^{T_f} C_{p,m} dT = \int_{T_i}^{T_f} \left( a + bT + \frac{c}{T^2} \right) dT = \left[ aT + \frac{b}{2}T^2 - \frac{c}{T} \right]_{T_i}^{T_f}$$

## Example 2B.2

$$\Delta H_m = \left( aT_f + \frac{b}{2}T_f^2 - \frac{c}{T_f} \right) - \left( aT_i + \frac{b}{2}T_i^2 - \frac{c}{T_i} \right)$$

$$T_i = 25^\circ\text{C} = 298\text{ K}, T_f = 100^\circ\text{C} = 373\text{ K}$$

Substitution leads to

$$\Delta H_m = H_m(373\text{ K}) - H_m(298\text{ K}) = 2.20\text{ kJ mol}^{-1}$$

# Changes in Enthalpy

For  $H = H(p, T)$ ,

$$dH = \underbrace{\left(\frac{\partial H}{\partial p}\right)_T}_{?} dp + \underbrace{\left(\frac{\partial H}{\partial T}\right)_p}_{C_p} dT$$

Let  $dH = 0$  (constant enthalpy).

$$\left(\frac{\partial H}{\partial p}\right)_T dp = -C_p dT$$

# Constant Enthalpy

At constant enthalpy,

$$\left(\frac{\partial H}{\partial p}\right)_T dp = -C_p dT$$

$$\left(\frac{\partial H}{\partial p}\right)_T = -C_p \left(\frac{\partial T}{\partial p}\right)_H$$

Define the **Joule-Thomson coefficient**:  $\mu = (\partial T / \partial p)_H$

$$\left(\frac{\partial H}{\partial p}\right)_T = -C_p \mu$$

# Changes in Enthalpy

For  $H = H(p, T)$ ,

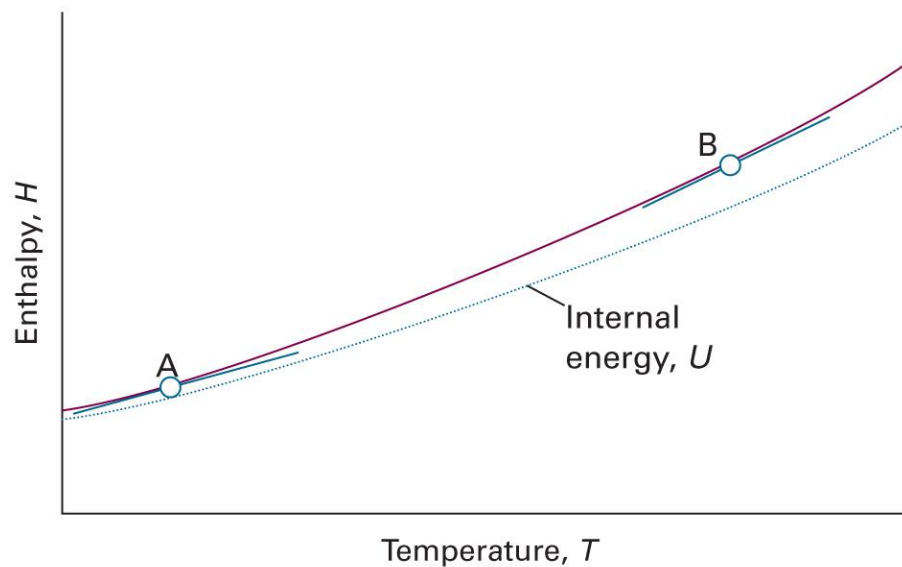
$$dH = \underbrace{\left(\frac{\partial H}{\partial p}\right)_T}_{-C_p\mu} dp + \underbrace{\left(\frac{\partial H}{\partial T}\right)_p}_{C_p} dT$$

$$dH = -C_p\mu dp + C_p dT$$

(Compare:  $dU = \pi_T dV + C_V dT$ )

# $C_V$ and $C_p$

$$C_V = \left( \frac{\partial U}{\partial T} \right)_V, \quad C_p = \left( \frac{\partial H}{\partial T} \right)_p$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11<sup>th</sup> ed., Figure 2B.3)

# Case of a Perfect Gas

$$C_p - C_V = \left( \frac{\partial H}{\partial T} \right)_p - \left( \frac{\partial U}{\partial T} \right)_p$$

For a perfect gas,

$$H = U + pV = U + nRT$$

so that

$$\begin{aligned} C_p - C_V &= \left( \frac{\partial (U + nRT)}{\partial T} \right)_p - \left( \frac{\partial U}{\partial T} \right)_p \\ &= \left( \frac{\partial U}{\partial T} \right)_p + nR - \left( \frac{\partial U}{\partial T} \right)_p = nR \end{aligned}$$

# Adiabatic Changes

$$\delta q = 0$$

$$dU = \delta q + \delta w = \delta w$$

“Adiabatic expansion”  $\rightarrow U$  drop  $\rightarrow T$  drop

# Work for Perfect Gas

$$dU = \delta w$$

For a perfect gas,  $dU = C_V dT$

$$C_V dT = \delta w$$

Integration leads to

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} C_V dT = \int_{T_i}^{T_f} C_V dT$$

If  $C_V$  is constant over  $[T_i, T_f]$ ,

$$w = C_V (T_f - T_i)$$

# Perfect Gas, Reversible Change

$$dU = \delta w$$

- Perfect gas:  $dU = C_V dT$
- Reversible change:  $\delta w = -pdV$

$$C_V dT = -pdV$$

Since  $p = nRT/V$ ,

$$\frac{C_V}{T} dT = -\frac{nR}{V} dV$$

# Integration

$$\int_{\text{state } i}^{\text{state } f} \frac{C_V}{T} dT = - \int_{\text{state } i}^{\text{state } f} \frac{nR}{V} dV$$

$$\int_{T_i}^{T_f} \frac{C_V}{T} dT = - \int_{V_i}^{V_f} \frac{nR}{V} dV$$

If  $C_V$  is constant over  $[T_i, T_f]$ ,

$$C_V \int_{T_i}^{T_f} \frac{1}{T} dT = -nR \int_{V_i}^{V_f} \frac{1}{V} dV$$

$$C_V (\ln T_f - \ln T_i) = -nR (\ln V_f - \ln V_i)$$

# Simplification

$$C_V(\ln T_f - \ln T_i) = -nR(\ln V_f - \ln V_i)$$

$$\frac{C_V}{nR} \ln \frac{T_f}{T_i} = -\ln \frac{V_f}{V_i} = \ln \frac{V_i}{V_f}$$

Let  $c = C_V/nR$ ;

$$\frac{C_V}{nR} \ln \frac{T_f}{T_i} = c \ln \frac{T_f}{T_i} = \ln \left( \frac{T_f}{T_i} \right)^c$$

$$\left( \frac{T_f}{T_i} \right)^c = \frac{V_i}{V_f}$$

# Volume-Temperature

For  $c = C_V/nR$ ,

$$\left(\frac{T_f}{T_i}\right)^c = \frac{V_i}{V_f}$$

Therefore,

$$V_i T_i^c = V_f T_f^c$$

or,

$$VT^c = \text{constant}$$

# From $T$ to $p$

$$V_i T_i^c = V_f T_f^c$$

For a perfect gas,  $T = pV/nR$

$$V_i \left( \frac{p_i V_i}{nR} \right)^c = V_f \left( \frac{p_f V_f}{nR} \right)^c$$

$$p_i^c V_i^{c+1} = p_f^c V_f^{c+1}$$

$$p_i V_i^{(c+1)/c} = p_f V_f^{(c+1)/c}$$

# Volume-Pressure

For  $c = C_V/nR$ ,

$$p_i V_i^{(c+1)/c} = p_f V_f^{(c+1)/c}$$

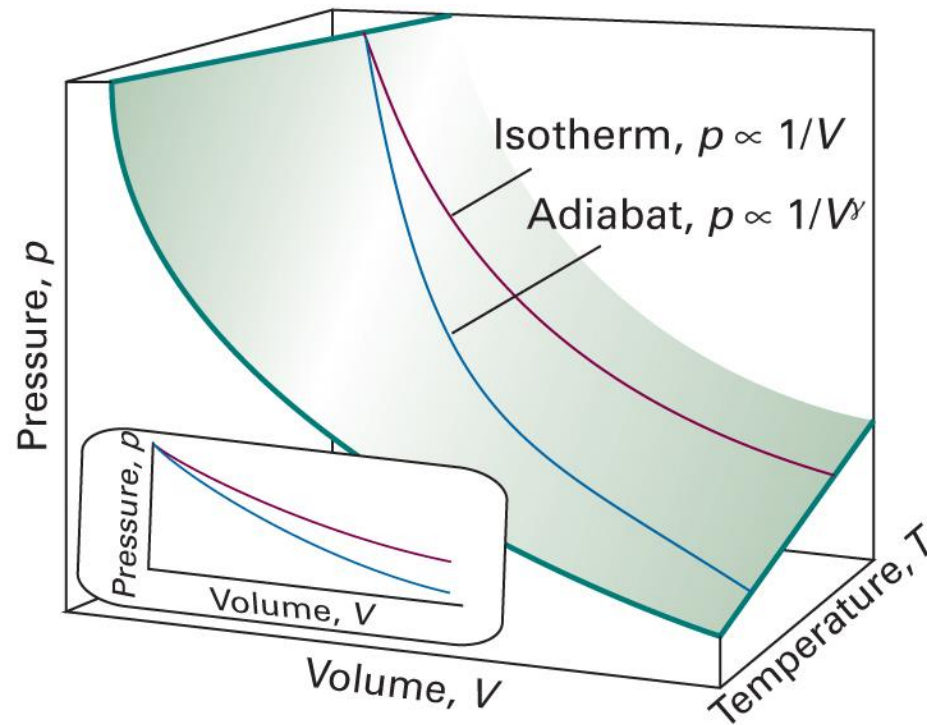
Let  $\gamma = (c + 1)/c = 1 + 1/c$ ;

$$p_i V_i^\gamma = p_f V_f^\gamma$$

or,

$$pV^\gamma = \text{constant}$$

# Isotherm and Adiabatic



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11<sup>th</sup> ed., Figure 2E.2)

# Monatomic Perfect Gas

$$VT^c = \text{constant}$$

$$pV^\gamma = \text{constant}$$

$$c = \frac{C_V}{nR} = \frac{(3/2)nR}{nR} = \frac{3}{2}$$

$$\gamma = 1 + \frac{1}{c} = 1 + \frac{2}{3} = \frac{5}{3}$$

# Summary for Chapter 3 (2)

| Thermodynamics | Mathematics          |
|----------------|----------------------|
| State function | Exact differential   |
| Path function  | Inexact differential |

Differential form of  $U$

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

Enthalpy

$$H = U + pV$$

$$dH = \delta q_p$$

Differential form of  $H$

$$dH = \left(\frac{\partial H}{\partial p}\right)_T dp + \left(\frac{\partial H}{\partial T}\right)_p dT$$

$$C_V = (\partial U / \partial T)_V$$

$C_V$  and  $C_p$

- Perfect gas
- Adiabatic change:  $\delta q = 0$

$$C_p = (\partial H / \partial T)_p$$